HATCHSOFT

LSC Learn Share Connect

Smart Academic Collaboration Platform

Task 1 Submission

Project Plan Site Maps and Wireframes

XISD5319 | Group 6

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Current GitHub repositories and workflow evidence

Website / PHP backend:https://github.com/MJjimmy/LSC

Android app:https://github.com/MJjimmy/LSC-APP

Website CI:https://github.com/MJjimmy/LSC/actions

Android CI:https://github.com/MJjimmy/LSC-APP/actions

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Project Plan

1 Project purpose objectives and scope

LSC Learn Share Connect is the current name of the StudySync platform in the original Task 1 submission. The project supports student and lecturer collaboration through a responsive website and an Android application using one PHP and MySQL backend on Railway. This plan retains the primary submission's team, requirements, schedule, budget, feasibility analysis and development approach, with current functionality reflected in the scope and navigation.

1.1 Purpose

This revision uses WILL PART 1 ASSIGNMENT.pdf as its primary source. StudySync is renamed LSC Learn Share Connect in this revised copy; the original PDF is unchanged. The original team, requirements, WBS, schedules, PERT, budget, assumptions, constraints, feasibility analysis and wireframes are retained or explicitly updated. Implementation evidence available on 19 October 2026 covers Android v1.3 (versionCode 4) and the existing Railway-hosted PHP/MySQL backend. Planned targets are not presented as completed tests or team approvals.

1.2 Project Overview

LSC LEARN SHARE CONNECT is an academic collaboration platform for students and lecturers. It brings together public/private study groups, membership requests and invitations, group chat, tasks, deadline calendar, shared files, notifications and profiles. Users access the shared PHP/MySQL service through a responsive website or an Android application, with administrator controls in a protected area.

1.3 Objectives

Objective	Success Measure
Deliver a consistent website and Android experience	Core features are accessible on desktop browsers, mobile browsers and the Android application.
Provide secure access to academic collaboration tools	Registration, login, session handling and role-based access pass functional and security testing.
Centralise group work and academic resources	Students can create/join groups, manage tasks, share files and plan deadlines from one account.
Maintain one reliable source of data	Website and Android application read and update the same MySQL records without duplication or inconsistency.
Apply a repeatable DevOps process	Every feature moves through planning, code review, build, test, release, deployment and monitoring.
Complete Task 1 documentation	Updated plan, site map and web/mobile wireframes are submitted in the required order.

Project Scope

1.4 Scope

- Responsive LSC LEARN SHARE CONNECT website for desktop and mobile browsers.
- Android Kotlin/XML application with a native welcome screen and a restricted HTTPS WebView for the hosted website.
- Student/lecturer registration, email/password and Google sign-in, password recovery and profile settings.
- Dashboard summaries for active groups, tasks, files and deadlines.
- Public/private study groups, approval-based join requests, invitations and group chat with attachments.
- Group task creation, self-assignment, pending/done status filtering and deadline tracking.
- Website group files/chat attachments support approved types up to 100 MB; profile avatars support approved image types up to 5 MB. Android v1.3 adds system file selection and authenticated saving, subject to client limits and device retesting.
- Monthly task-deadline calendar and in-app notifications for invitations and membership decisions.
- Protected administrator dashboard with system totals and activation/deactivation of other accounts.
- Shared PHP application, MySQL 8, Railway deployment, private file storage, responsive PWA support and GitHub Actions.
- Automated and manual testing, deployment documentation, monitoring and release records.

Out of Scope

- Integration with external university learning-management or student-information systems.
- Advanced artificial-intelligence tutoring or automated academic content generation.
- Native iOS application development.
- Complex offline synchronisation and background file transfers.
- Commercial payment, subscription or advertising features.

High Level Requirements

ID	Functional Requirement	Priority
FR 1	Student/lecturer registration, email/password and Google sign-in, password recovery.	High
FR 2	Public/private groups, approval-based requests and recipient-bound invitations.	High
FR 3	Permitted group files and chat attachments up to the displayed 100 MB limit.	High
FR 4	Create self-assigned group tasks, filter by status/group and mark own task done.	High
FR 5	View monthly deadlines for tasks in joined groups.	Medium
FR 6	In-app invitation and membership notifications; email subject to preferences and queue processing.	Medium
FR 7	Protected administrator totals and activation/deactivation of other accounts.	Low
FR 8	Group text chat and attachments shared with the file library.	High
FR 9	Android native welcome, restricted WebView, Back navigation and error recovery.	High

ID	Non Functional Requirement	Target
NFR 1	System uptime	99% during peak hours
NFR 2	Page load time	Under 3 seconds
NFR 3	Concurrent users	50 or more simultaneous users
NFR 4	Browser support	Chrome, Firefox and Edge
NFR 5	Security	Hashed passwords, HTTPS, CSRF protection and permission checks

The availability, performance and concurrency figures are acceptance targets from the primary plan; they require measured test evidence.

Important Implementation Changes

Android v1.3 adds system document-picker uploads and authenticated downloads to the existing Kotlin/XML welcome-and-WebView client. It supports explicitly granted content URIs, up to 10 chosen files with a 100 MB combined client limit, a system save-location picker and completed-stage preservation while that picker is recreated. Google sign-in uses the system browser and a one-time verifier-bound return. Its reported browser-session rejection remains open; deployment of diagnostics is not a confirmed fix. Installation and full physical-device testing of v1.3 have not been confirmed.

The website supports profiles, light/dark appearance and a PWA offline fallback. It does not synchronise collaboration data offline. Tasks cannot currently be edited or reassigned; the calendar is monthly. Chat refreshes periodically. Email uses Resend; saved SMS preferences do not establish live SMS delivery.

Deliverables and Team Allocation

The following deliverables and owners are planning allocations retained from the primary submission. Role descriptions are aligned with the cover titles; actual contributions, meeting attendance and approvals require separate evidence.

1.5 Deliverables

ID	Deliverable	Acceptance Evidence	Owner(s)
D1	Updated project plan	Scope, schedule, allocations, DevOps lifecycle, risks and success measures documented.	Seabelo / All
D2	Site map	Role-based website and mobile navigation structure shown and explained.	Keabetswe / Gladwin
D3	Updated wireframes	Important website and Android screens represented with navigation and input controls.	Gladwin / Keabetswe
D4	Development repository and workflow	GitHub branches, issues, pull requests, coding standards and release tags configured.	Seabelo / Gladwin
D5	Responsive website increment	Authentication, groups, tasks, files, calendar and admin pages integrated.	Gladwin
D6	Android mobile application	Kotlin WebView app launches securely, loads the hosted system and supports main navigation.	Mothoka
D7	Connected database and migrations	MySQL schema, constraints, backup procedure and migration scripts verified.	Gladwin
D8	Testing and release package	Test cases, defect log, UAT evidence, deployment evidence and final documentation.	Keabetswe / All

1.6 Group Responsibilities and Deliverables

Member	Responsibilities	Main Deliverables
Seabelo Blessing Motloung Project Manager / DevOps Coordinator	Maintain scope, schedule, backlog, meetings, risks and change control. Configure the GitHub workflow, coordinate releases, confirm deployment readiness and compile the final document.	Updated plan; sprint board; status reports; release checklist; deployment record; final submission.
Keabetswe Ramahlape Malwa Business Analyst / QA Coordinator	Update requirements and acceptance criteria, validate the site map and wireframes, create test cases, coordinate usability testing and manage the defect log.	Requirements updates; acceptance criteria; traceability; test plan; UAT results; defect summary.
Gladwin Engetani Nkuna Database Administrator	Maintain the MySQL schema, constraints and migrations; review database permissions, prepared queries, backups and recovery; support Railway database configuration and data-consistency tests. This is the planned DBA allocation, not proof of work performed.	Wireframes; website modules; Android build; integration code; unit tests; technical notes.
Mothoka Malatji Developer / UI-UX Designer	Produce website and mobile interface designs, implement PHP/UI integration and the Kotlin/XML WebView client, write code-	Schema and migrations; backup plan; database test evidence; security checks; data dictionary updates.

Member	Responsibilities	Main Deliverables
	level tests and resolve interface defects. This is the planned Developer/UI-UX allocation, not proof of work performed.	

Work Breakdown and Schedule

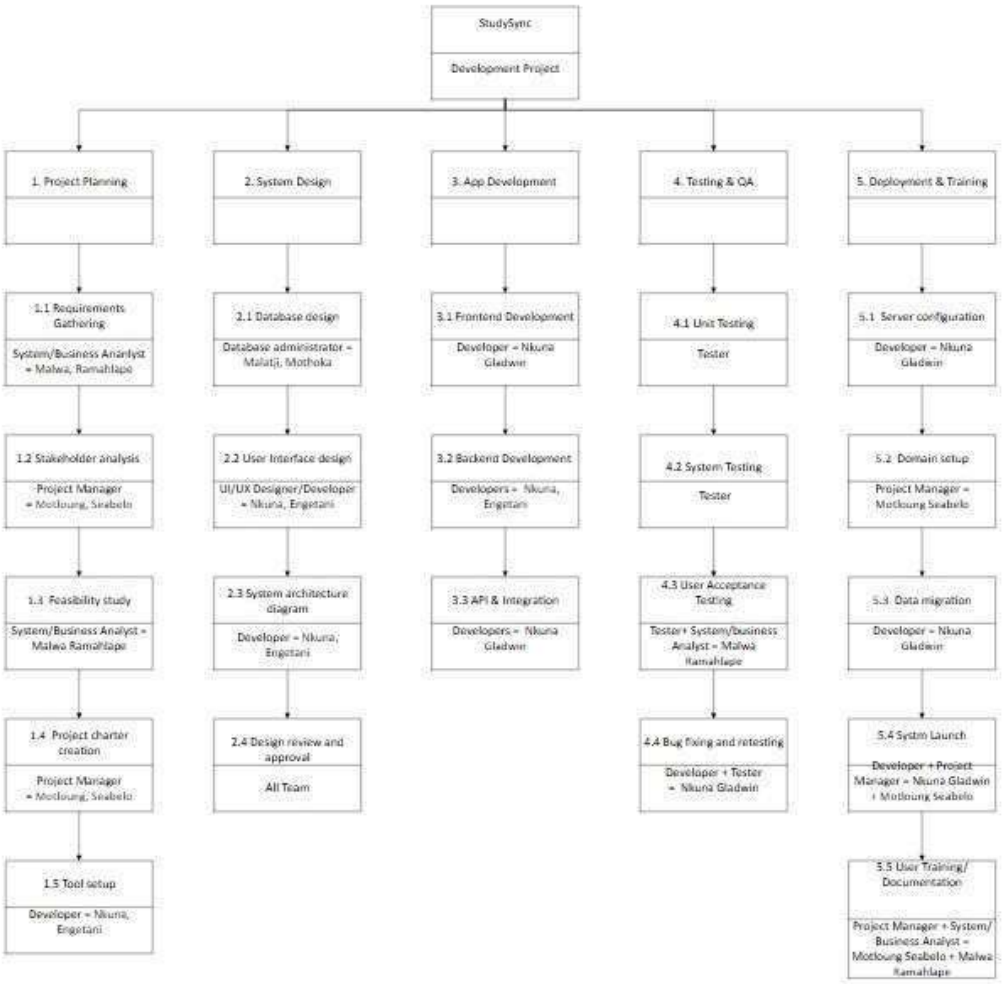


Figure 1. Work breakdown structure retained from the primary submission.

1.7 Work Breakdown Interpretation

The work breakdown separates planning, design, implementation, testing and delivery into reviewable work packages. Website, Android and database work share the same account and collaboration model. Each work package should have an owner, acceptance criteria and evidence in GitHub.

Period	Primary Plan Workstream
Weeks 1 to 2	Planning and analysis
Weeks 3 to 4	System design
Weeks 5 to 7	Development
Week 8	Testing
Week 9	Deployment
Week 10	Documentation and final submission

The detailed schedule figures retain the original planning baseline. Android integration and the current feature set must be included in the development and test work packages.

Detailed Delivery Schedule

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		Project Initiation	5 days	26/04/21	26/04/25	
2		Create Project Charter	3 days	26/04/21	26/04/23	
3		Define Objectives & Scope	2 days	26/04/24	26/04/25	2
4		Requirements Phase	5 days	26/04/28	26/05/02	
5		Gather Functional Requirements	3 days	26/04/28	26/04/30	
6		Gather Non-Functional Requirements	2 days	26/05/01	26/05/02	5
7		Requirements Sign-off	0 days	26/05/04	26/05/04	6
8		System Design Phase	10 days	26/05/04	26/05/15	7
9		System Architecture Design	3 days	26/05/05	26/05/07	
10		UI/UX Wireframes & Mockups	4 days	26/05/05	26/05/08	
11		Database Schema Design	3 days	26/05/11	26/05/13	10
12		Design Review & Approval	2 days	26/05/14	26/05/15	11
13		Design Complete	1 day	26/05/18	26/05/18	12
14		Development Phase 1	10 days	26/05/19	26/05/30	13
15		User Registration & Logi	5 days	26/05/19	26/05/23	
16		Study Group Creation Module	5 days	26/05/26	26/05/30	
17		Join/Manage Groups Feature	3 days	26/05/26	26/05/28	16
18		Dev Phase 1 Complete	1 day	26/05/30	26/05/30	17
19		Development Phase 2	10 days	26/06/02	26/06/13	

Figure 2. Delivery schedule part one from the primary submission.

The first schedule segment breaks planning, requirements, design and initial implementation into dependencies. Each successor starts after the relevant prerequisite or approved parallel activity.

Detailed Delivery Schedule Continued

	Task Mode	Task Name	Duration	Start	Finish	Predecessors
20		Task Management System	4 days	26/06/02	26/06/05	
21		File Upload & Sharing	4 days	26/06/06	26/06/10	20
22		Calendar & Scheduling	3 days	26/06/11	26/06/13	21
23		Notification System	3 days	26/06/11	26/06/13	22
24		Dev Phase 2 Complete	0 days	26/06/15	26/06/15	23
25		Testing Phase	7 days	26/06/16	26/06/24	24
26		Unit Testing	2 days	26/06/16	26/06/17	
27		Integration Testing	2 days	26/06/18	26/06/19	25
28		System Testing & Performance	3 days	26/06/18	26/06/22	26
29		Bug Fixing & Regression	1 day	26/06/24	26/06/24	27
30		Testing Complete	1 day	26/06/24	26/06/24	28
31		Deployment Phase	5 days	26/06/25	26/07/01	29
32		Server Configuration	2 days	26/06/25	26/06/26	
33		Database Migration	1 day	26/06/27	26/06/27	30
34		Final Deployment & Launch	2 days	26/06/30	26/07/01	31
35		Launch Complete	0 days	26/07/01	26/07/01	32
36		Project Closure	5 days	26/07/02	26/07/08	33
37		Final Documentation	3 days	26/07/02	26/07/04	
38		User Manual & Training	1 day	26/07/07	26/07/07	34
39		Post-Launch Review	1 day	26/07/08	26/07/08	35
40		Project Closure	0 days	26/07/08	26/07/08	36

Figure 3. Delivery schedule part two from the primary submission.

Delivery Schedule Gantt Views

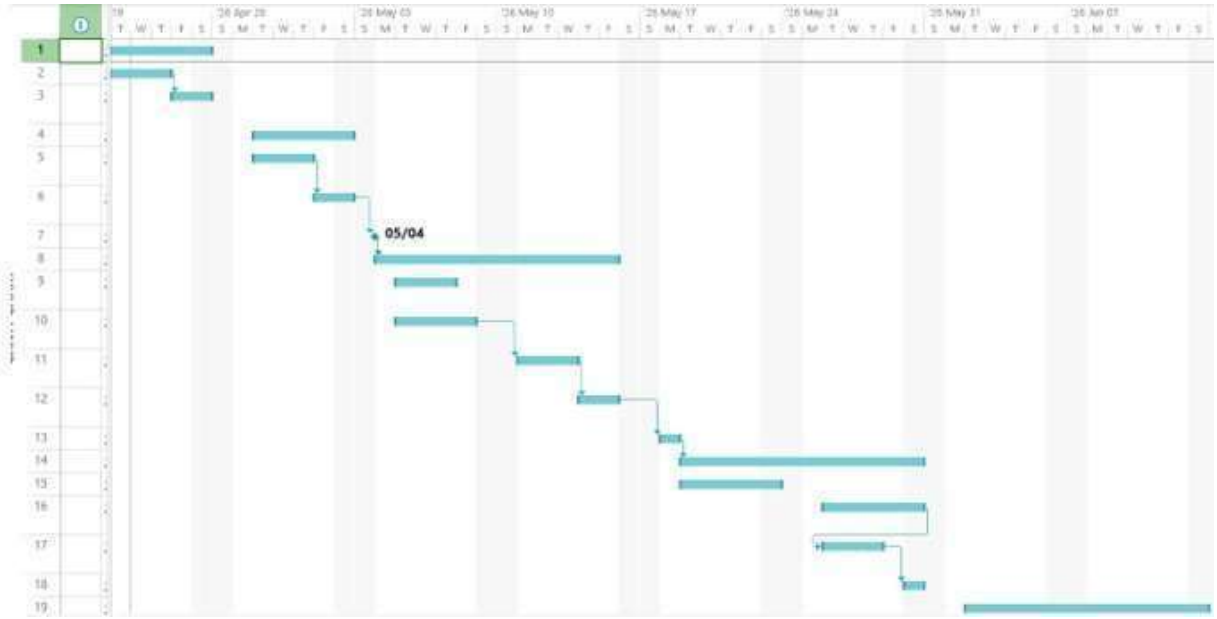


Figure 4. Delivery schedule Gantt view part one from the primary submission.

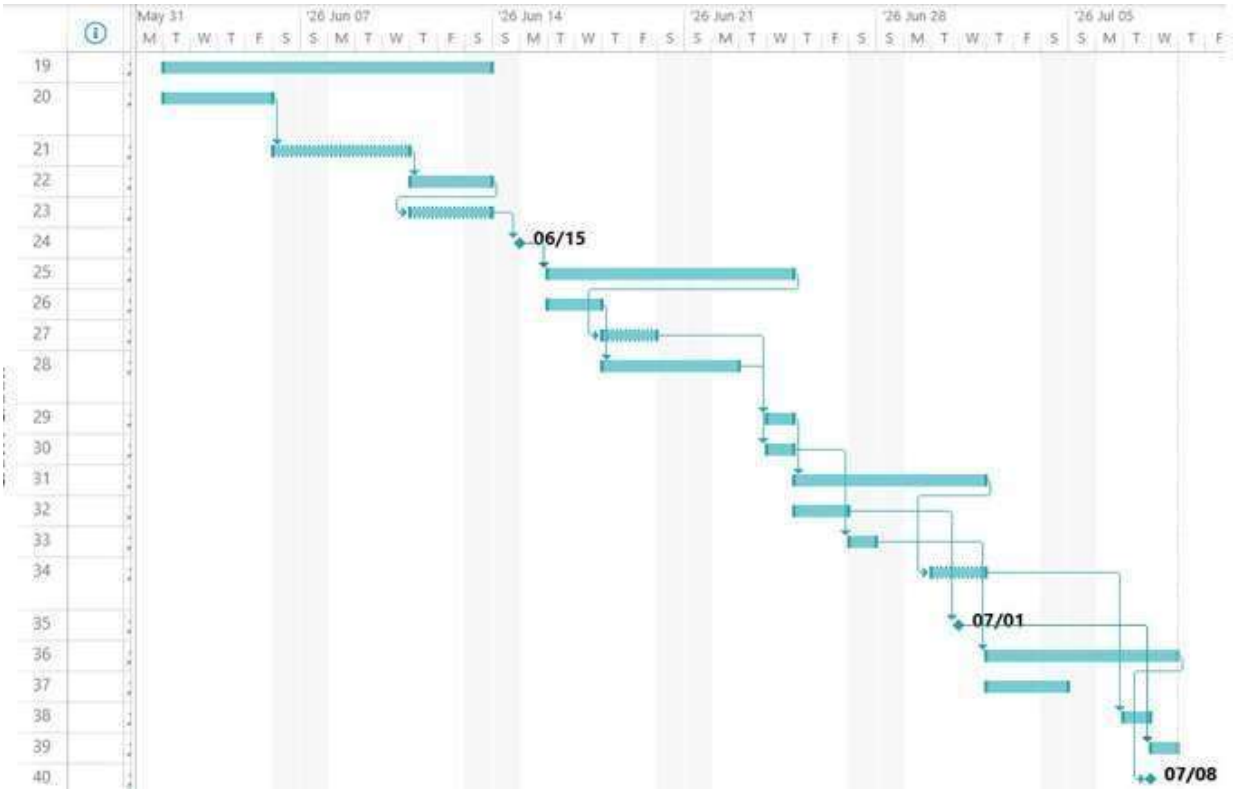


Figure 4 continued. Delivery schedule Gantt view part two from the primary submission.

Milestones PERT and Budget

1.8 Milestones and Dependencies

The summary milestone targets below are a planning sequence, not actual completion dates. The detailed primary schedule is retained in Figures 2–4, and its dates must be reconciled with the PERT estimate.

Milestone	Target	Dependency
Kick off and roles	Week 1	None
Requirements approved	Week 2	Kick off
Design complete	Week 4	Requirements
Authentication and groups complete	Week 5	Design
Tasks and files complete	Week 7	Phase one
Testing complete	Week 8	Phase two
Launch	Week 9	Testing
Project closure	Week 10	Launch and documentation

The original dated schedule and Gantt figures are retained as a historical planning baseline, not as verified completion dates. Reconcile their dates with the 10-week PERT estimate and milestone targets before using them for a revised release commitment.

PERT Activities and Interpretation

Activity	Description	Duration	Predecessor
A	Planning and requirements	2 weeks	None
B	System design	2 weeks	A
C	Combined development phases	4 weeks	B
D	Testing	1.5 weeks	C
E	Deployment	0.5 weeks	D
F	Documentation	Throughout	A

The PERT sequence A → B → C → D → E totals 10 weeks. A delay to any sequential activity delays completion unless time is recovered. Documentation runs alongside delivery after planning begins. The primary milestone table and PERT estimate place release at different points; the team must reconcile that baseline before committing to release dates.

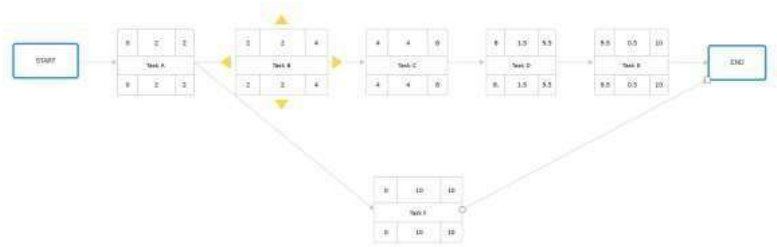


Figure 5. PERT network from the primary submission.

Budget Estimate

Category	Item	Estimate	Basis
Hosting	Cloud server for three months	R300	Planning allowance for Railway; confirm actual charges
Domain	One year	R300	Optional domain allowance
Tools	Project management, GitHub and design	R0	Free or institution-provided access

Category	Item	Estimate	Basis
Contingency	Unexpected costs	R200	Reserve from the primary plan
Total	Project budget	R800	Estimate rather than confirmed expenditure

Assumptions Constraints and Risks

Assumptions

- Members have reliable internet and suitable devices.
- Each member can dedicate 5 to 8 hours per week and has basic web development knowledge.
- Required tools are available free or through the university.
- Users have basic computer literacy and modern browsers.
- Requirements remain stable unless a change is approved.
- Hosting remains within the estimate and the university network permits access.
- The work fits within a 10 week academic term alongside other coursework.

Constraints

- Fixed 10 week deadline, R800 budget and four team members.
- Use free or open-source tools where possible.
- The scope includes the Android Kotlin WebView app alongside the website; the old exclusion of mobile development is superseded.
- Some technologies require team learning; exams and coursework affect availability.
- No dedicated server infrastructure initially; browser compatibility must be maintained.

Risk Register

Category	Risk	Probability	Impact	Score
Technical	Implementation difficulties	Medium	High	High
Technical	Server downtime	Medium	High	High
Technical	Data loss or corruption	Low	High	Medium
Technical	Browser compatibility	Medium	Medium	Medium
Team	Delays and missed deadlines	Medium	High	High
Team	Member leaves	Low	High	Medium
Team	Communication breakdown	Medium	Medium	Medium
Project	Requirement changes	Medium	Medium	Medium
Project	Scope creep	High	Medium	High
External	Data security breach	Low	High	Medium
External	Competing deadlines	High	Medium	High
External	Internet or infrastructure failure	Low	Medium	Low

Risk Management

Risk	Mitigation	Contingency
Technical difficulties	Regular tests, reviews and peer work	Simplify lower priority features
Server downtime	Reliable hosting and verified backups	Restore service using a tested recovery plan
Data loss	Plan regular backups and test restores	Restore the last verified backup
Browser issues	Test multiple browsers early	Prioritise supported browsers and fix defects
Team delays	Clear ownership and weekly check-ins	Adjust scope and lower priority work
Member leaves	Cross-train critical tasks	Redistribute responsibilities
Communication issues	Weekly meetings and shared documents	Daily check-ins when blocked
Requirement changes	Document requirements and approve changes	Defer optional changes
Scope creep	Change control and stakeholder agreement	Record future phase requests
Security breach	Secure authentication and permission checks	Patch promptly and notify affected users as appropriate
Competing deadlines	Plan around exams and start early	Replan realistically

Backup, restore and recovery measures are planned controls; the guide does not establish that a fully automatic backup programme is operating.

Technical and Economic Feasibility

Technical Feasibility and Skill Assessment

Skill	Availability	Gap	Response
HTML and CSS	High	None	Use existing skills
JavaScript	Medium	Advanced concepts	Tutorials and peer learning
PHP backend	Low to medium	Framework and server experience	Start simply and review code
Database design	Medium	Optimisation	Focus on sound schema and queries
Security	Low	Authentication and safe configuration	Use established patterns and review
Deployment	Low	Server configuration	Document Railway deployment
Android Kotlin and WebView	Requires validation	Native lifecycle and browser handoff	Build, test and verify on devices

Technology Stack Justification

Component	Current Technology	Reason
Web interface	HTML, CSS and JavaScript	Responsive shared interface
Server	PHP controllers, services and repositories	Reuses the implemented platform
Database	PDO and MySQL 8	Shared records and parameterised queries
Android	Kotlin/XML with restricted WebView	Native entry and recovery with shared web features
Hosting	Railway with nginx and PHP FPM	Hosted backend and persistent private upload storage
Version control and builds	GitHub and GitHub Actions	Reviewable changes, checks and Android artifacts
Email	Resend	Password reset and queued invitation email

Economic Feasibility

Cost Type	Amount or Status	Notes
Development labour	R0	Student team without wages
Software licences	R0 planned	Free/open-source or institution access
Domain	R300 allowance	Confirm need and actual annual price
Hosting for three months	R300 allowance	Confirm runtime, database and volume charges
Maintenance after project	Unconfirmed	Requires a continuing owner and budget
Initial cash budget	R800 planned	Includes R200 contingency; actual expenditure unverified

DevOps and Release Workflow

1.9 DevOps Lifecycle and Application

LSC LEARN SHARE CONNECT will use a continuous DevOps lifecycle rather than treating development, testing and deployment as separate one-time activities. Each small feature will move through the same controlled stages. This gives the group early feedback, prevents last-minute integration problems and creates clear evidence of individual contributions.



Figure 6. Primary DevOps lifecycle retained as a planning diagram; current delivery uses GitHub/GitHub Actions and Railway.

Stage	How It Will Be Applied to LSC LEARN SHARE CONNECT	Primary Evidence
Plan	Convert requirements into prioritised GitHub issues with acceptance criteria, owners, estimates and sprint targets.	Backlog, sprint plan, issue assignments.
Code	Use feature branches. Developers commit small changes, follow naming standards and link commits to issues.	Branch history, commits, coding checklist.
Build	Run automated syntax checks and package the website and Android application. Build failures block release.	GitHub Actions results, Android build output.
Test	Execute unit, integration, database, responsive-interface, security and user-acceptance tests.	Test results, defect log, screenshots.
Release	Merge approved changes into the relevant source/deployment branch after checks. The current Android source branch is	Pull-request approval, release tag, notes.

Stage	How It Will Be Applied to LSC LEARN SHARE CONNECT	Primary Evidence
	master; the PHP production branch is deploy/railpack. Release tags and team approvals require their own evidence.	
Deploy	Deploy the web release to Railway and create the Android release build that points to the hosted HTTPS address.	Railway deployment log, APK/build evidence.
Operate	Maintain user access, database availability, file storage and basic support procedures.	Operational checklist, backup confirmation.
Monitor	Review application errors, deployment health, database issues and tester feedback; create new issues for improvements.	Logs, uptime checks, feedback issues.

1.10 GitHub and Release Workflow

1. Create or update a GitHub issue for every feature, defect or documentation change.
2. Create a branch using a consistent name such as feature/groups-search or fix/login-validation.
3. Commit logical changes with descriptive messages and link them to the issue.
4. Open a pull request and require at least one group member to review the change.
5. Run automated checks and complete the relevant manual tests before merging.
6. Promote only tested changes to Android master or PHP deploy/railpack. Do not overwrite the separate preserved main histories.
7. Deploy the approved version to Railway, build the Android application and record the release evidence.
8. Monitor the release and return any new defect or improvement to the backlog.

Quality Communication and Success Measures

1.11 Quality Management

Quality Area	Control
Requirements quality	Every feature has a clear description, acceptance criteria, priority and owner.
Code quality	Peer review, consistent naming, reusable components, prepared statements and input validation.
Interface quality	Responsive layouts, consistent navigation, readable contrast, clear empty states and mobile touch targets.
Database quality	Primary/foreign keys, validation constraints, migration scripts, backups and data-consistency tests.
Testing quality	Unit, integration, system, responsive, security and UAT coverage with traceable results.
Release quality	No open critical defects, successful build, approved checklist and rollback/backup readiness.

1.12 Risk and Change Management

Risk	Likelihood / Impact	Mitigation and Owner
Limited time for both platforms	High / High	Reuse the shared hosted website through Kotlin WebView, prioritise core requirements and track weekly milestones. Owner: Seabelo.
Mobile and website session inconsistency	Medium / High	Use HTTPS, test login/logout across platforms and document cookie/session behaviour. Owner: Gladwin.
Database migration or data loss	Medium / High	Use versioned migration scripts, test on a copy and confirm backups before deployment. Owner: Mothoka.
Uneven contribution or missed work	Medium / Medium	Assign issues and deliverables by member, review status weekly and escalate blockers early. Owner: Seabelo.
Interface does not match requirements	Medium / Medium	Review sitemap and wireframes with the Business Analyst before implementation. Owner: Keabetswe.
Deployment failure near submission	Medium / High	Use incremental Railway deployments, keep a known stable release and prepare rollback steps. Owner: Seabelo / Mothoka.

1.13 Communication and Reporting

- Weekly planning meeting to confirm completed work, next priorities, risks and dependencies.
- Short progress updates in the group communication channel at least twice per week.
- GitHub issues and pull requests as the official record of technical work and review.
- One shared document folder for plans, diagrams, test evidence and submission files.
- Decisions affecting scope, architecture or deadlines recorded in a change log.

1.14 Success Measures

Measure	Target
Documentation completeness	All required headings are present in the exact order and all figures

Measure	Target
	are labelled.
Core feature readiness	All high-priority functions have approved wireframes, acceptance criteria and assigned work items.
Cross-platform consistency	The website and Android app use the same account, groups, tasks, files and calendar data.
Security readiness	HTTPS, hashed passwords, validated input, access controls and secure database queries are tested.
Measure	Target
Quality threshold	No critical defects remain open before release; all agreed UAT scenarios pass.
Deployment readiness	A repeatable GitHub-to-Railway workflow and Android build process are documented.

Final acceptance should record website and device tests, Google handoff and session behaviour, permission boundaries, Android file handling and actual database checks. User-reported phone tests and repository checks do not independently verify every edge case.

Current Implementation and Evidence — 19 October 2026

This section records project evidence, not a blanket completion certificate. Original budget, timing and non-functional figures remain planning targets. The current repositories were checked against the recorded release commits during this revision.

Area	Current implementation	Evidence / remaining work
Android release	v1.3 / code 4; Kotlin/XML, native Welcome and shared HTTPS WebView. File/photo selection and authenticated save workflow implemented.	28 JVM tests passed locally and in GitHub Actions run 39258572727; device picker/permission and upload/download retesting outstanding.
PHP website	Same PHP 8.4 controllers/services/repositories and MySQL 8.4 service on Railway; persistent private upload volume.	47 disposable HTTP checks passed in run 39256382794; source ab756da. No production Google sign-in or inbox-delivery pass claimed.
Google SSO	System-browser OAuth, session-bound confirmation and verifier-bound native exchange. Extra Profile Google button removed.	Open issue: missing/changed browser-session state causes rejected confirmations. Boolean-only diagnostics deployed; no security check was disabled.
Groups and files	Approval-based public joining; creator/Admin group deletion requires confirmation and CSRF. Web files 100 MB; avatars 5 MB.	Older Android build lacked picker/dialog support. v1.3 includes the client handler; the owner has not confirmed installation or device retest.
Messaging and settings	Group chat/attachments; account profile, notification preferences and web-page light/dark preference.	Resend worker handles eligible queued email subject to preferences. Actual recipient inbox delivery and SMS delivery are not established.
DevOps evidence	Android source: MJjimmy/LSC-APP master at 9c03104. PHP deploy source: MJjimmy/LSC deploy/railpack at ab756da.	Lecturer permission to use GitHub instead of Azure DevOps is reported by the owner. Keep written approval; Issues/Projects, team meetings and peer evaluations remain unverified.
Operational limits	PWA provides a limited offline fallback, not offline collaboration replay. Tasks are self-assigned; calendar is monthly.	Do not claim editing/reassignment, weekly calendar, general event creation, audit reports or full offline synchronisation as implemented.
Final submission	Retain this primary-plan revision, current source, test/build records and user-support documentation.	A narrated feature demonstration and current physical-phone evidence are still required; measured performance and budget targets are not verified outcomes.

Database Population Snapshot

The table below reproduces the project owner's successful read-only MySQL console output supplied on 19 October 2026. It was not remeasured during document editing. Server version reported by that console: MySQL 8.4.11. A demo SQL file was prepared, but no committed inserts have been confirmed.

Table	Records	Interpretation
app_auth_codes	1	Authentication/security data: do not invent identities or usable tokens. Confirm the record-target interpretation with the lecturer.
files	2	Needs 8 actual file uploads to reach 10; do not insert nonexistent file paths.
group_invitations	4	Below 10 in the supplied snapshot; additional legitimate/demo records not yet confirmed.
group_join_requests	2	Below 10 in the supplied snapshot; additional legitimate/demo records not yet confirmed.
group_members	12	At least 10 in the supplied snapshot.
messages	2	Below 10 in the supplied snapshot; additional legitimate/demo records not yet confirmed.
notification_queue	11	At least 10 in the supplied snapshot.
notifications	20	At least 10 in the supplied snapshot.
oauth_identities	2	Authentication/security data: do not invent identities or usable tokens. Confirm the record-target interpretation with the lecturer.
password_reset_tokens	1	Authentication/security data: do not invent identities or usable tokens. Confirm the record-target interpretation with the lecturer.

Table	Records	Interpretation
rate_limits	105	At least 10 in the supplied snapshot.
study_groups	7	Below 10 in the supplied snapshot; additional legitimate/demo records not yet confirmed.
tasks	3	Below 10 in the supplied snapshot; additional legitimate/demo records not yet confirmed.
users	8	Below 10 in the supplied snapshot; additional legitimate/demo records not yet confirmed.

Current evidence references

Android: source commit 9c031041bccc9186dc4072b725f90daaf4ebf108; GitHub Actions 39258572727; 28 tests, no failures/errors/skips. Backend diagnostics: ab756daa58b18d2d7d94c2fe0c6b7252ccb3f592; GitHub Actions 39256382794; Railway deployment 6379a226-3d3d-4a63-aecf-5537fe3989b6. These identify the checked artifacts; neither is independent physical-phone or provider-delivery evidence.

Website Site Map

The website uses a persistent global navigation structure after sign-in, with a nested group workspace. The arrows show entry routes and transitions. They indicate navigation; server-side checks still govern protected actions.

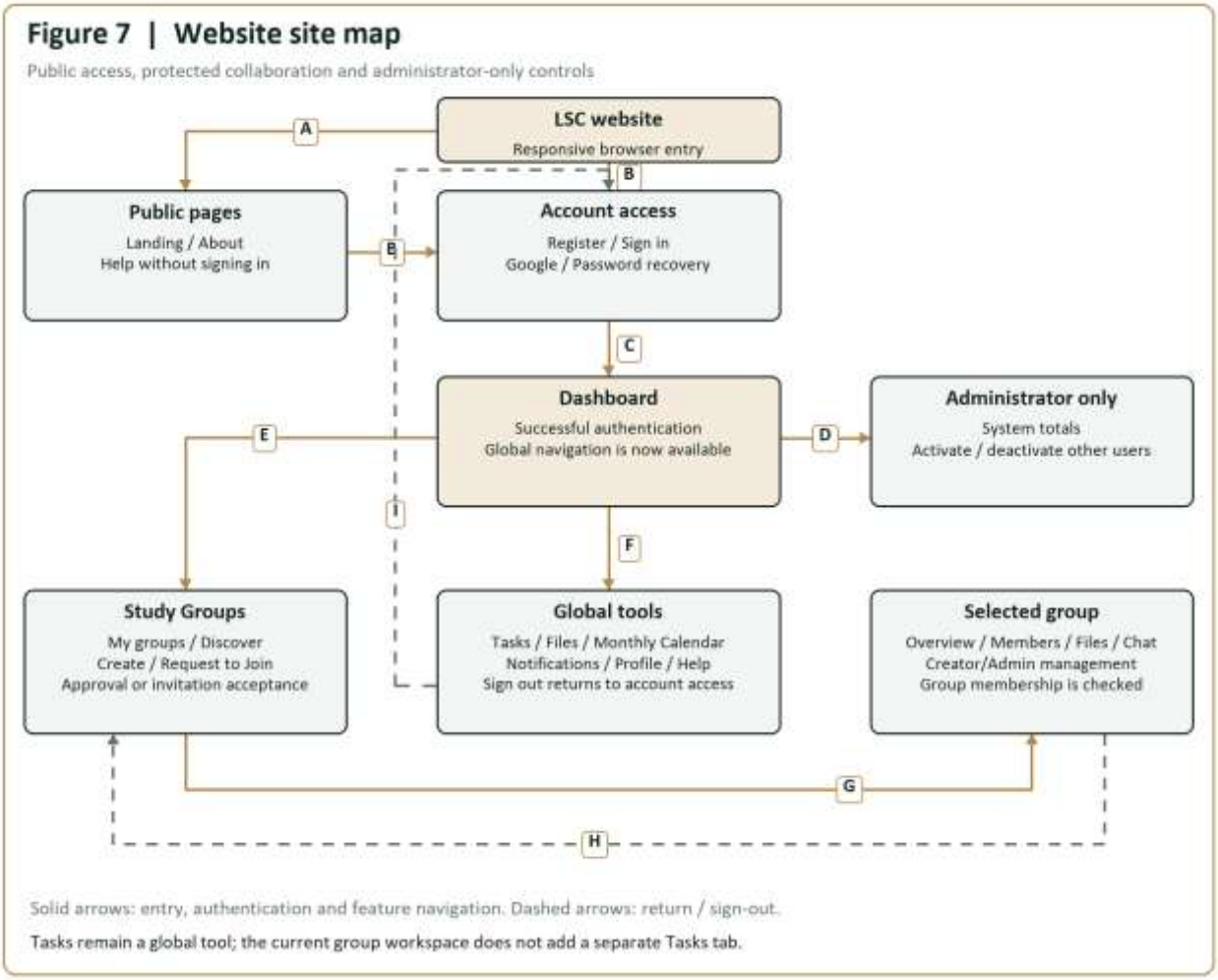


Figure 7. Website site map with labelled navigation paths.

Website Navigation Patterns

Figure 7 path	Navigation pattern and feedback
A	Browser entry opens the public Landing/About area or public Help. These paths do not grant access to collaboration records.
B	Register creates a Student/Lecturer account and returns to Sign in. Sign in offers email/password or Google; password recovery returns to account access. Failed validation stays in the access flow.
C	Successful authentication establishes the PHP session and opens Dashboard. A protected URL without a valid active session returns to account access.
D	Only a current Admin role exposes/permits the protected system-totals and account-activation controls. This branch is not available to ordinary Student/Lecturer accounts.
E	Dashboard/global navigation opens Study Groups. My Groups opens memberships; Discover sends a public-group Request to Join; Create Group creates the group and creator membership.
F	Global navigation switches directly between Tasks, Files, monthly Calendar, Notifications, Profile and Help. Users need not visit these tools in a fixed sequence. On narrow screens, the shared responsive bottom navigation exposes Home, Groups, Tasks, Files and Calendar.
G	Opening an authorised selected group enters Overview, Members, Files or Chat while preserving that group ID. The current workspace does not add a separate Tasks tab; use global Tasks and its group filter. Creator/Admin controls approve requests, invite/manage members or delete a group where authorised.
H	The return path leads back to Groups without logging out. Switching workspace tabs retains the selected group; changing the global tool changes the page context.
I	Sign out explicitly ends the session and returns to account access. Browser Back is not a substitute for Sign out; protected requests are rechecked by the server.

Desktop screens keep a left sidebar. Narrow web screens use responsive navigation and the same feature labels. The map combines hierarchical navigation into groups with global movement between tools; it is not a sequence that requires users to visit every tool.

Access and Feedback Rules

The group creator or administrator may manage requests and members where authorised. File downloads independently require actual membership, including for administrators. Only the uploader can delete a file. Failed access, empty lists and validation errors provide feedback within the relevant page.

Android App Site Map

Android adds native entry and recovery around the same hosted web application. This map separates native screens, the trusted WebView and the system-browser Google flow, making the app navigation different from the website map.

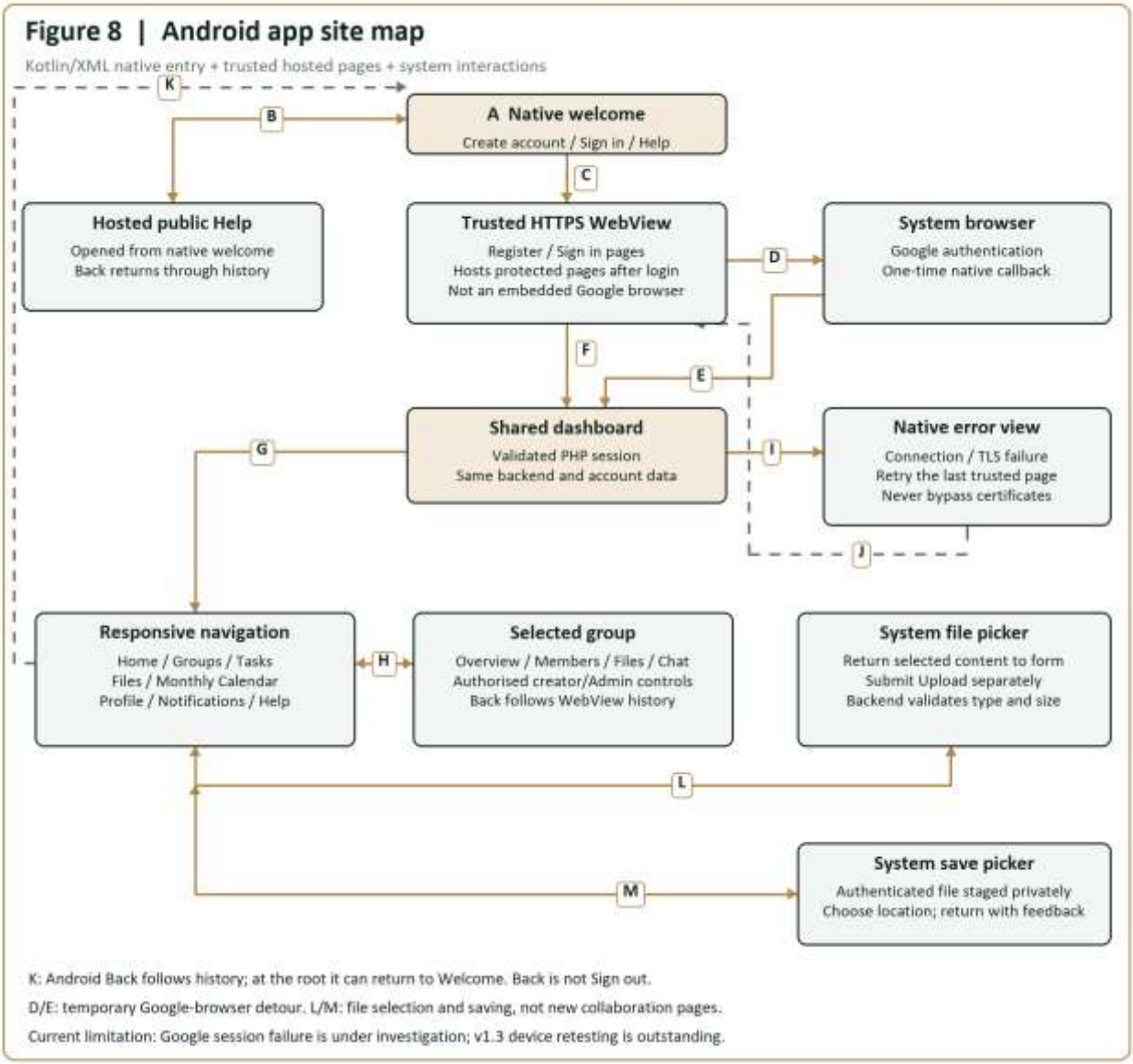


Figure 8. Android app site map with labelled navigation paths.

Android Navigation Patterns

Figure 8 path	Navigation pattern and feedback
A	App launch shows the animated native Welcome screen. Its entry choices are Create an account, Sign in and Help; the app does not require the public website landing page first.
B	Help opens the hosted public support page. Back follows the previous page/native entry path; Help remains reachable without authentication.
C	Create an account or Sign in loads the corresponding trusted HTTPS page in WebView. Registration success returns to account access rather than bypassing authentication.
D	Continue with Google is intercepted by Android and opens the system browser. Google authentication is not embedded inside WebView.
E	After verified provider completion, the browser returns a one-time code plus native state. Android checks its pending state and redeems the code with its private verifier before opening the authenticated WebView session. Missing/expired/mismatched state is rejected; the reported phone-session failure remains open.
F	Valid email/password authentication, or a successfully redeemed Google return, opens the shared Dashboard. Session cookies remain subject to server expiry and active-account checks.
G	The hosted responsive navigation switches between Home, Groups, Tasks, Files and monthly Calendar. Profile, Notifications and Help are shared web destinations, not separately rebuilt native feature screens.
H	An authorised selected group uses Overview, Members, Files and Chat; no separate Tasks tab is claimed. Group filters in global Tasks/Files preserve the selected academic context. Return navigation follows WebView history.
I	Connection/TLS failures show the native error view rather than an empty page or certificate bypass. This is different from an OAuth confirmation rejection shown by the browser.
J	Retry reloads the last trusted URL. It does not guarantee Dashboard navigation, accept an arbitrary host or ignore a TLS failure.
K	Android Back traverses WebView history; at its root it can return to native Welcome. Back does not clear the account session. Explicit Sign out is still required.
L	From a file/photo input in Files, Chat or Profile, Android opens the system document picker. Returning a permitted content URI selects the file on the same form; the user still submits Upload. Canceling returns no selection. Server membership/type/size checks remain authoritative.
M	A direct authenticated same-origin HTTPS download is staged privately, then the system save picker lets the user choose a destination. Success/cancellation returns feedback to the app. Completed picker stages survive configuration recreation; interrupted network/copy operations are canceled rather than resumed. Device verification remains outstanding.

The WebView rejects cleartext/mixed content and keeps app features on the trusted backend. D/E are a temporary system-browser detour, while L/M are native document-selection/save interactions. The file pipeline is implemented in v1.3, but passing JVM policy/wiring tests is not evidence of every Android provider, multipart upload or device-lifecycle case.

Wireframes

3.1 Wireframe Purpose

The updated wireframes translate the current requirements and site map into clear page structures for the website and mobile application. They focus on information hierarchy, navigation, inputs, actions and system feedback. Colour, imagery and final branding can be refined during implementation, but the placement and purpose of the main controls should remain stable unless a reviewed change is approved.

The primary submission's wireframes are retained as design references. LSC Learn Share Connect replaces the earlier StudySync project name. Current requirements and annotations below define the implemented behaviour where older design controls suggest broader functionality.

3.2 Shared Interface Principles

Principle	Application in the Wireframes
Consistency	The same labels, icons and main navigation areas appear across web and mobile.
Clear hierarchy	Page title, primary action, filters and main content are visually separated.
Responsive behaviour	Desktop side navigation becomes a mobile menu or bottom navigation.
Accessible interaction	Readable text, labelled inputs, visible focus/error states and large touch targets.
Feedback	Loading, success, validation, empty and error states are planned for every data action.
Role awareness	Student, group-owner and administrator controls are shown only where appropriate.

Each screen explanation describes how a user enters the screen, the actions available and the return route. The website and Android maps define the overall navigation; the wireframes provide the screen layouts within those paths.

Website Wireframes

3.3 Authentication and Dashboard

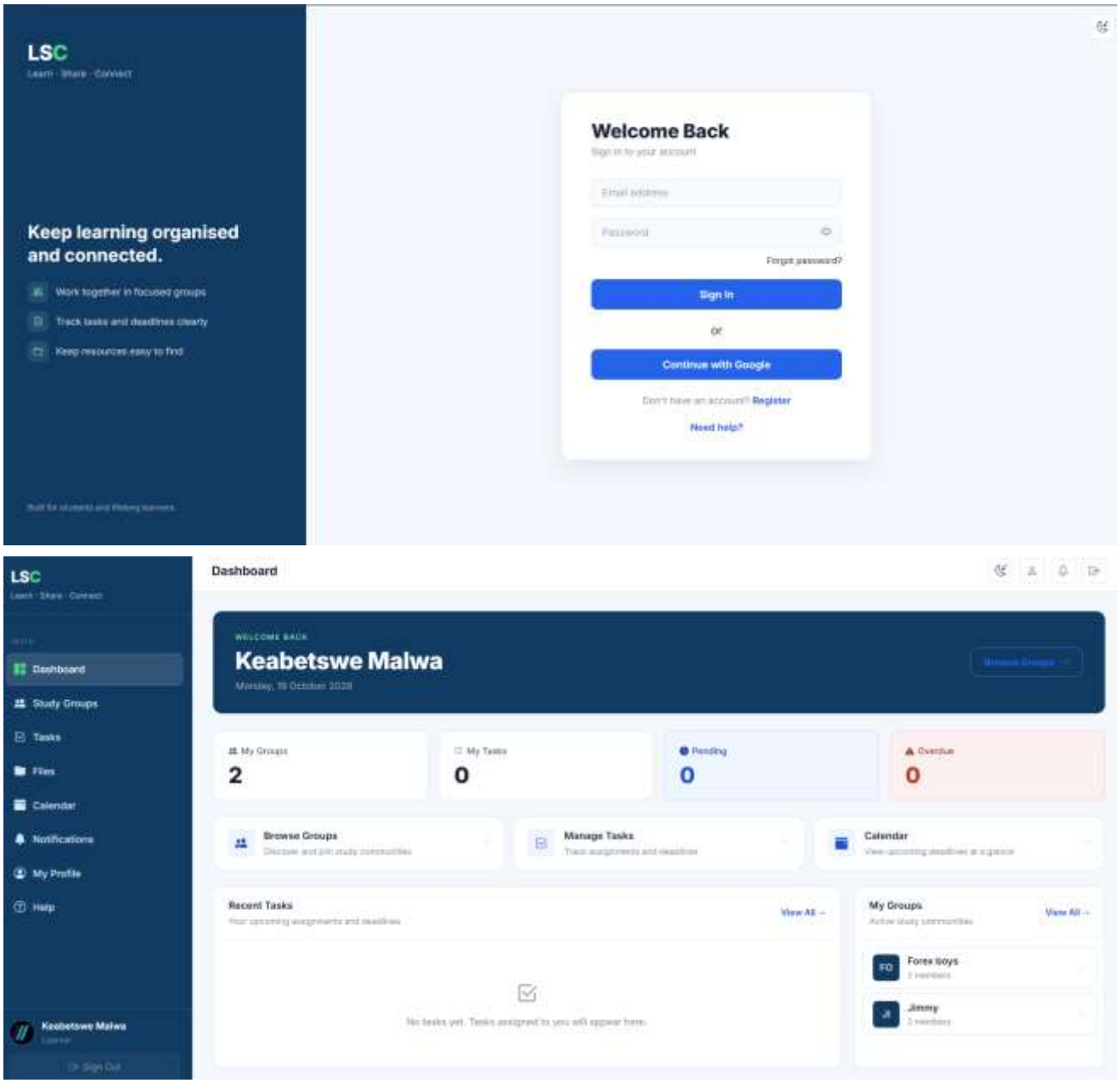


Figure 9. Primary website wireframes for authentication and dashboard.

From the landing page, Register opens account creation and Sign in opens authentication. Password recovery returns to sign-in after reset. Successful authentication opens Dashboard, whose shortcuts and sidebar lead to Groups, Tasks and Calendar. Current account access also supports Lecturer registration and Google sign-in. Dashboard task totals are assigned to the current user.

Study Groups and Workspace

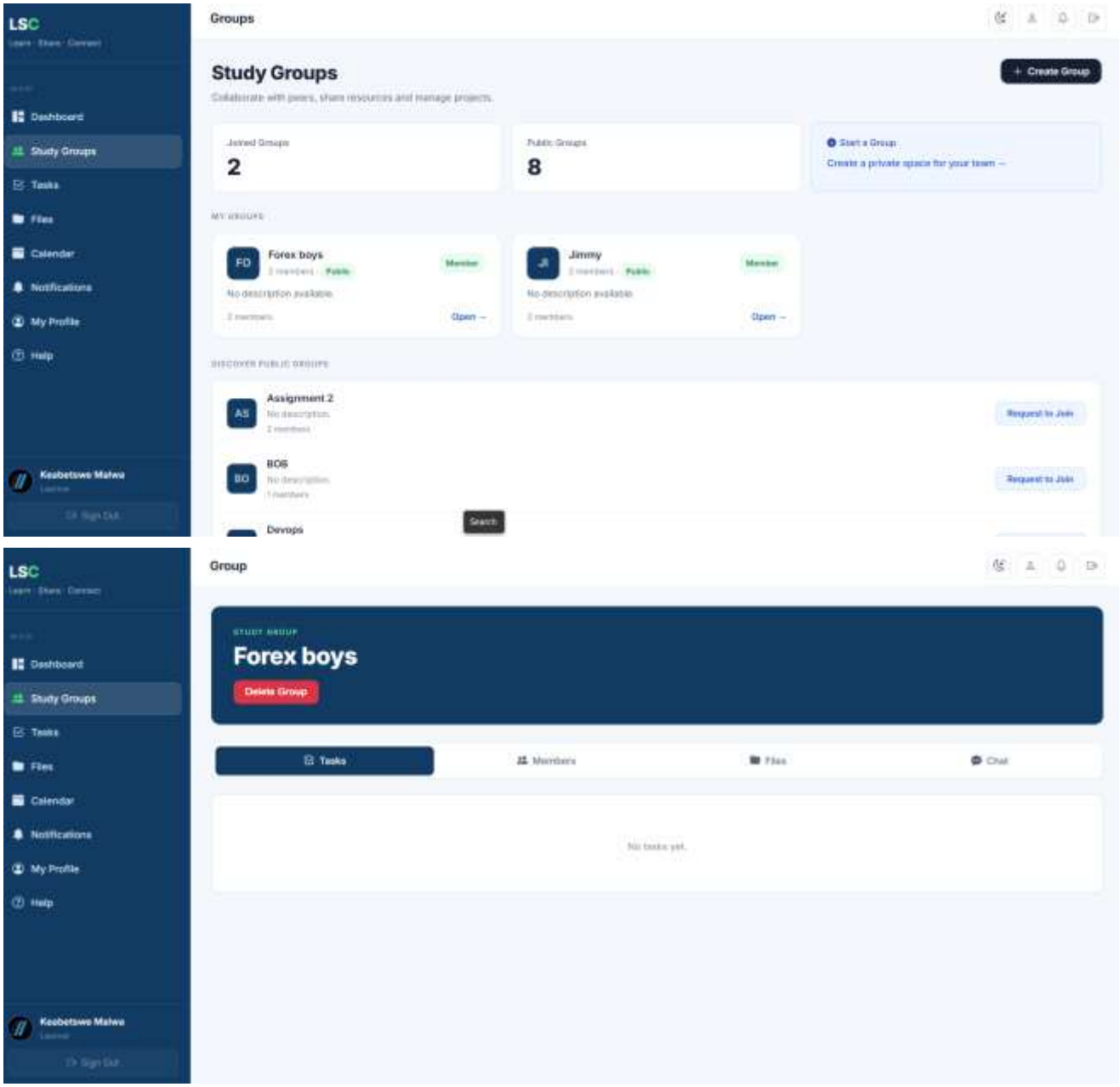


Figure 10. Primary website wireframes for groups and workspace.

My Groups opens a selected workspace. Discover sends a Request to Join and displays a pending state; Create Group opens its creation form. Overview, Members, Files and Chat retain the selected group ID. Tasks opens through global navigation and a group filter, not a current group Tasks tab. Groups returns to the list. Creator/Admin actions remain permission checked, and Delete Group first requires a confirmation and valid CSRF submission.

Website Tasks Files and Administration

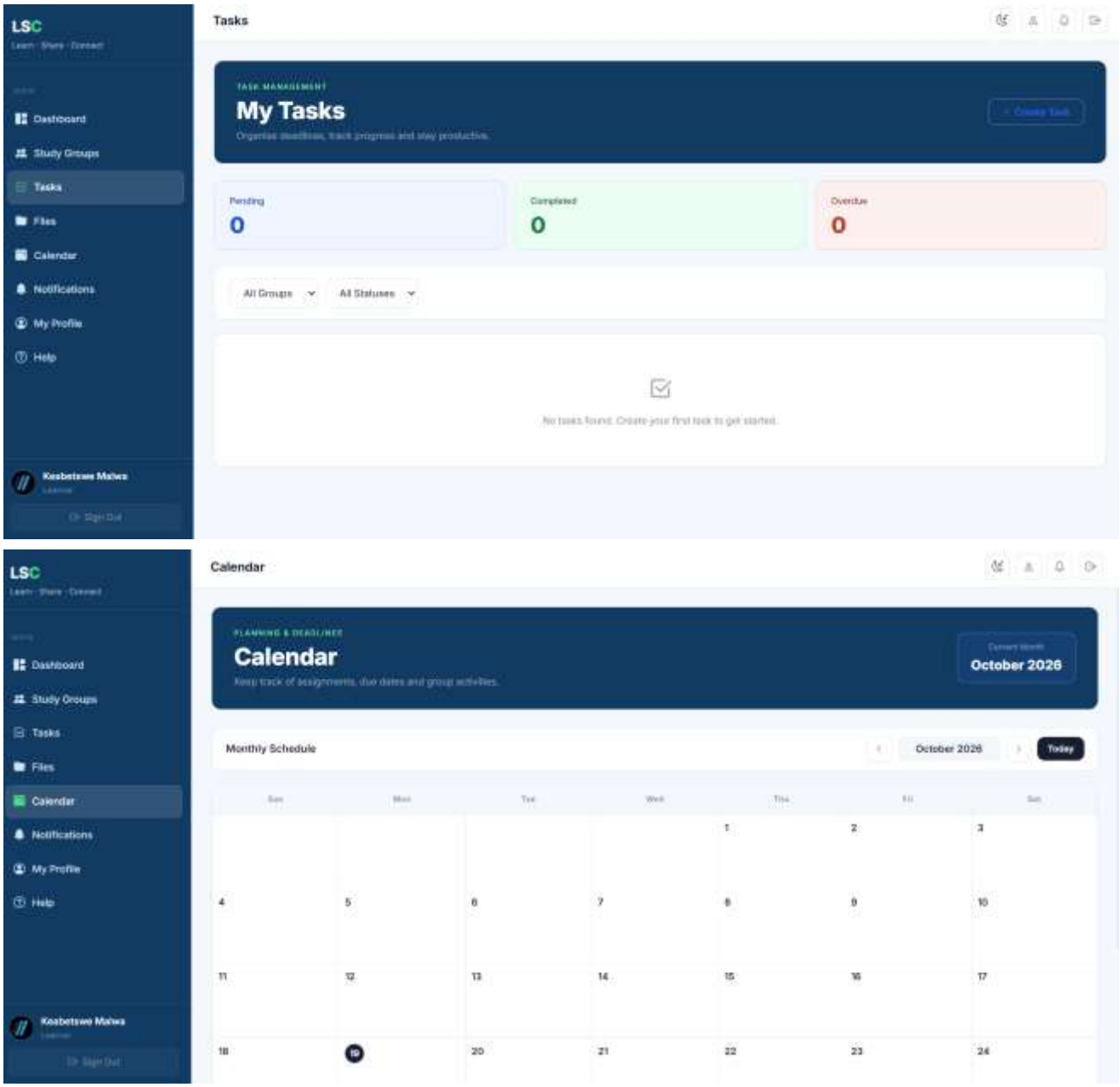
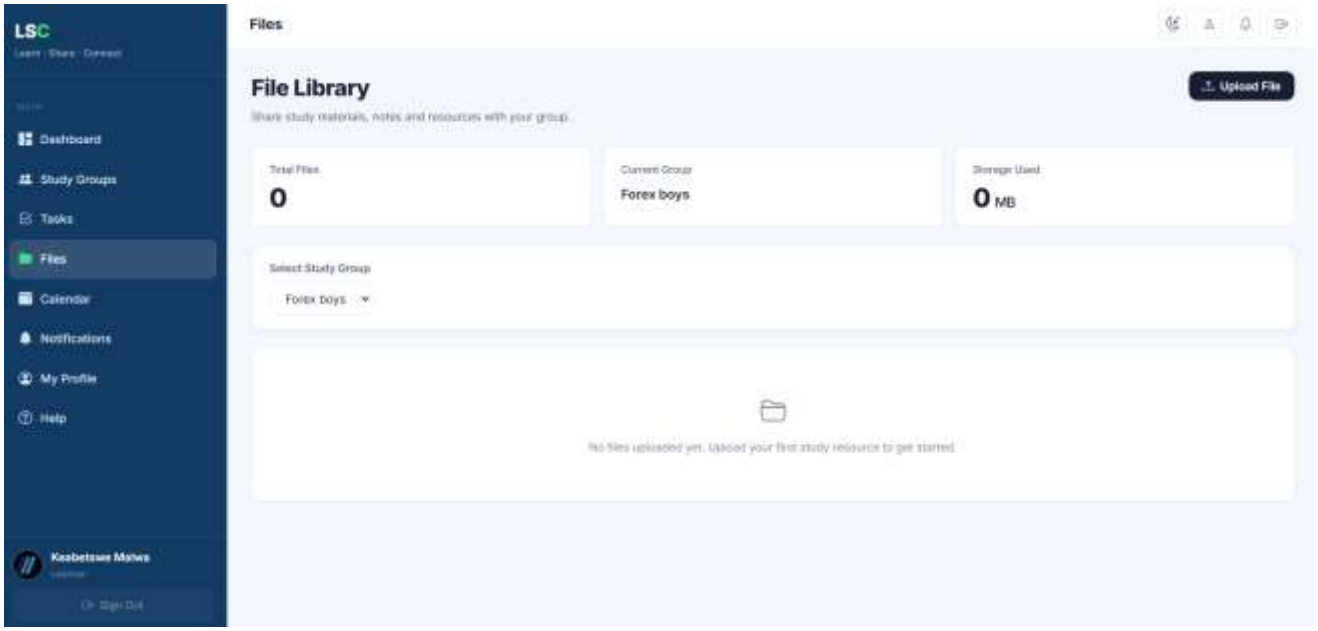
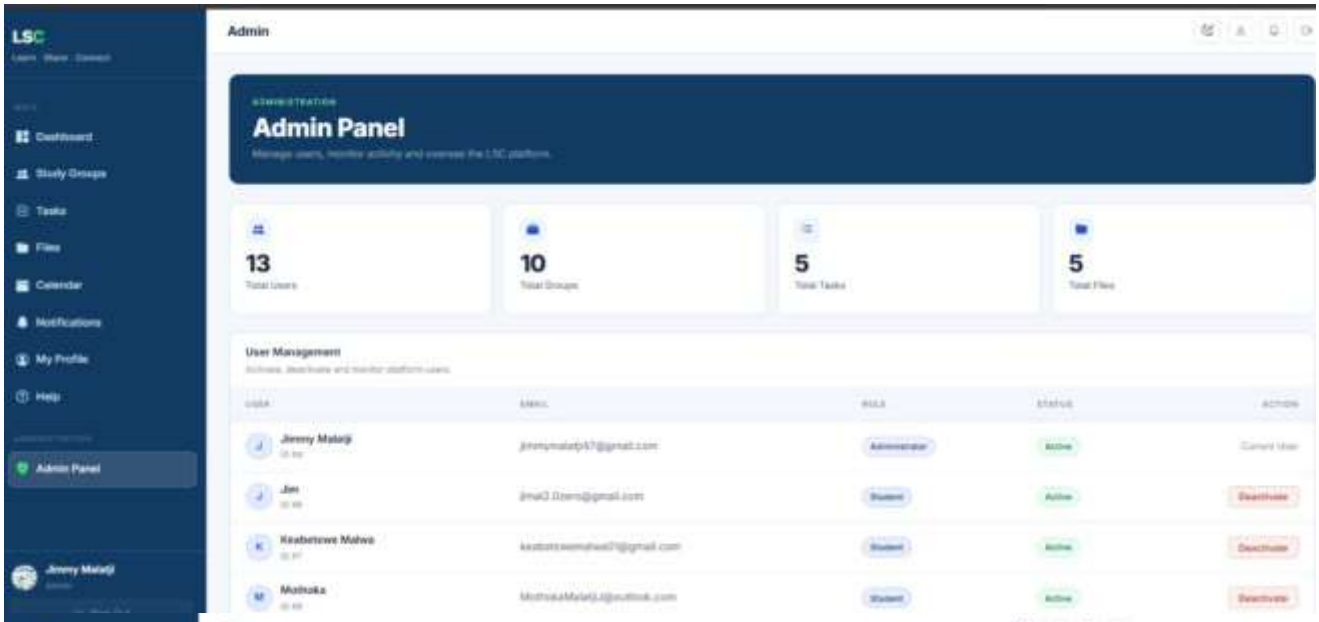


Figure 11. Primary website wireframes for tasks and calendar.

Tasks opens from global navigation; group/status filters narrow the list. Create Task assigns the task to its creator, and the assignee can mark it done. Calendar switches to a monthly view derived from joined-group task due dates. Its previous/next-month controls change the displayed period. Older edit/reassignment, weekly-view or Add Event controls in the retained planning images are not claimed as implemented.


LSC
Learn · Share · Connect
Dashboard
Study Groups
Tasks
Files
Calendar
Notifications
My Profile
Help
Keabetswe Malwa
Files
File Library
Share study materials, notes and resources with your group.
Upload File
Total Files
0
Current Group
Forex boys
Storage Used
0 MB
Select Study Group
Forex Boys
No files uploaded yet. Upload your first study resource to get started.


LSC
Learn · Share · Connect
Dashboard
Study Groups
Tasks
Files
Calendar
Notifications
My Profile
Help
Admin Panel
Admin
ADMINISTRATION
Admin Panel
Manage users, monitor activity and oversee the LSC platform.
13
Total Users
10
Total Groups
5
Total Tasks
5
Total Files
User Management
Activate, deactivate and monitor platform users.
USER
EMAIL
ROLE
STATUS
ACTION
Jimmy Malatji
Administrator
Active
Current User
Jon
Student
Active
Deactivate
Keabetswe Malwa
Student
Active
Deactivate
Mothoka
Student
Active
Deactivate

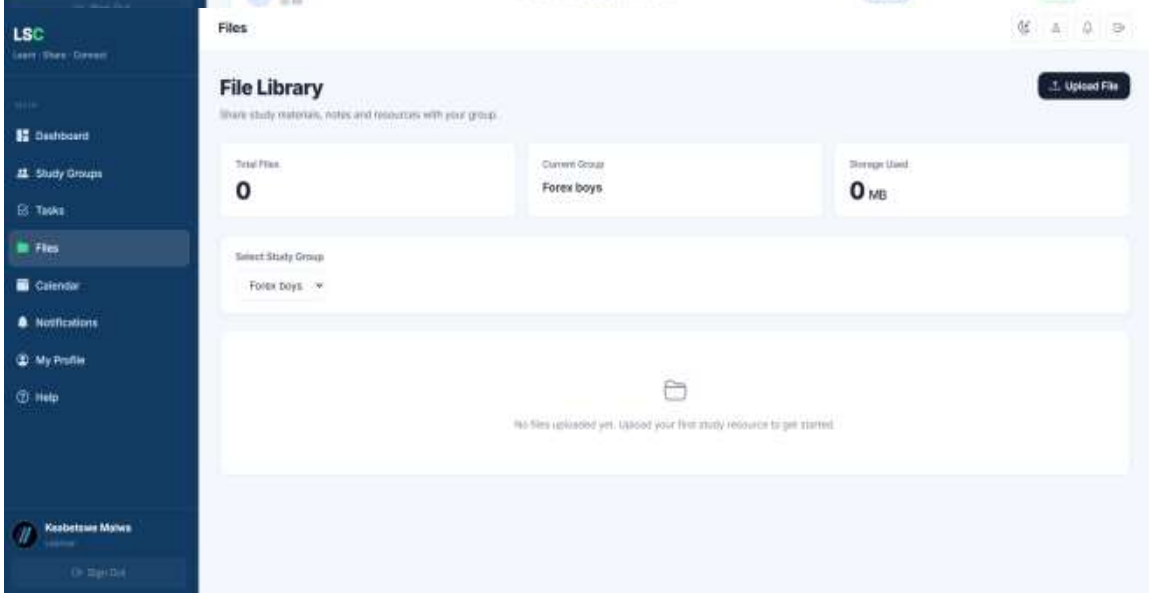
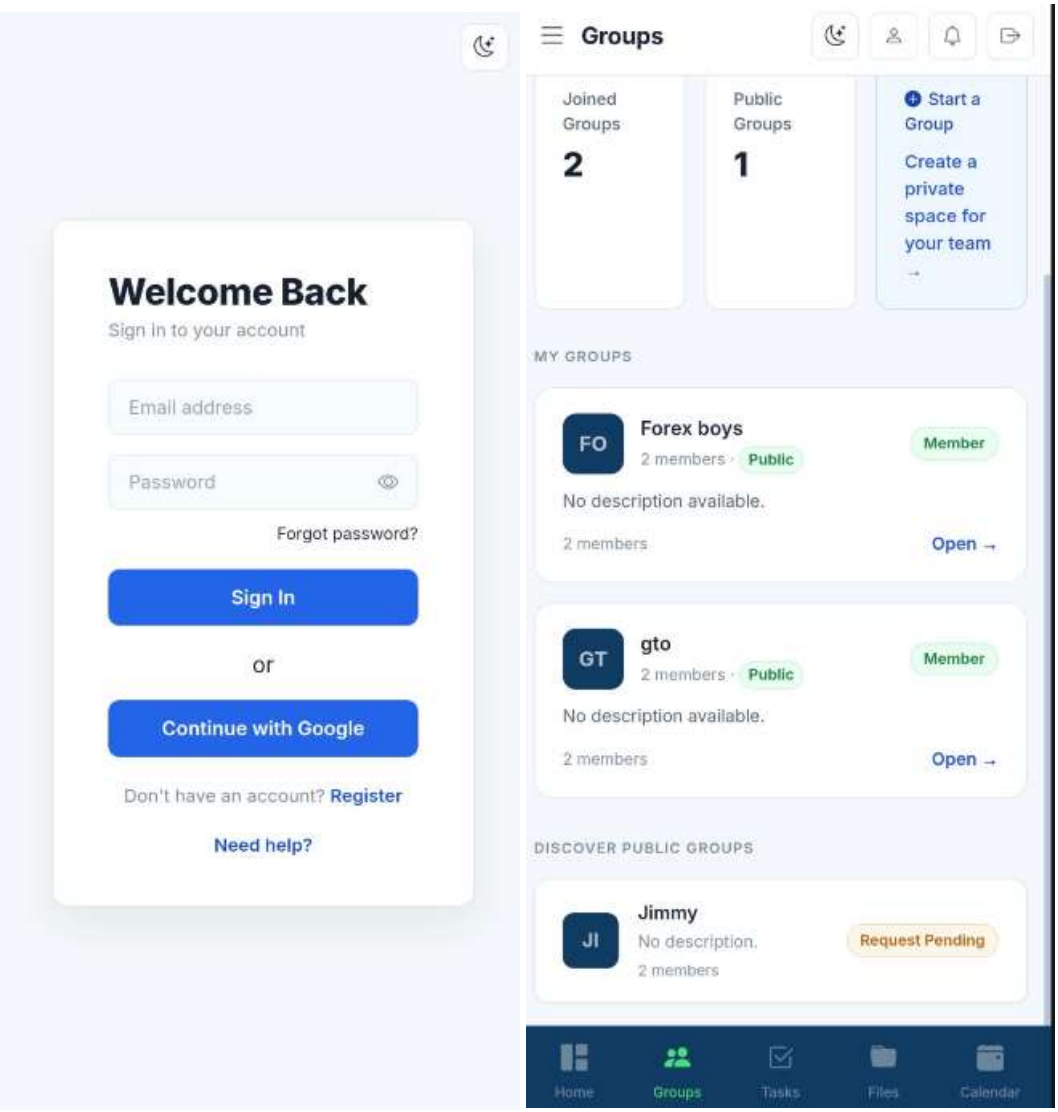

LSC
Learn · Share · Connect
Dashboard
Study Groups
Tasks
Files
Calendar
Notifications
My Profile
Help
Keabetswe Malwa
Files
File Library
Share study materials, notes and resources with your group.
Upload File
Total Files
0
Current Group
Forex boys
Storage Used
0 MB
Select Study Group
Forex Boys
No files uploaded yet. Upload your first study resource to get started.

Figure 12. Primary website wireframes for file library and administration.

Files opens a group-filtered library. Upload requires a joined group, an allowed type and the displayed 100 MB limit; authenticated download checks membership. Only uploaders delete their files. Admin is a protected destination for totals and activation/deactivation of other accounts; the current plan does not claim role editing or audit reports as implemented screens.

Mobile Application Wireframes

3.4 Login Home and Groups



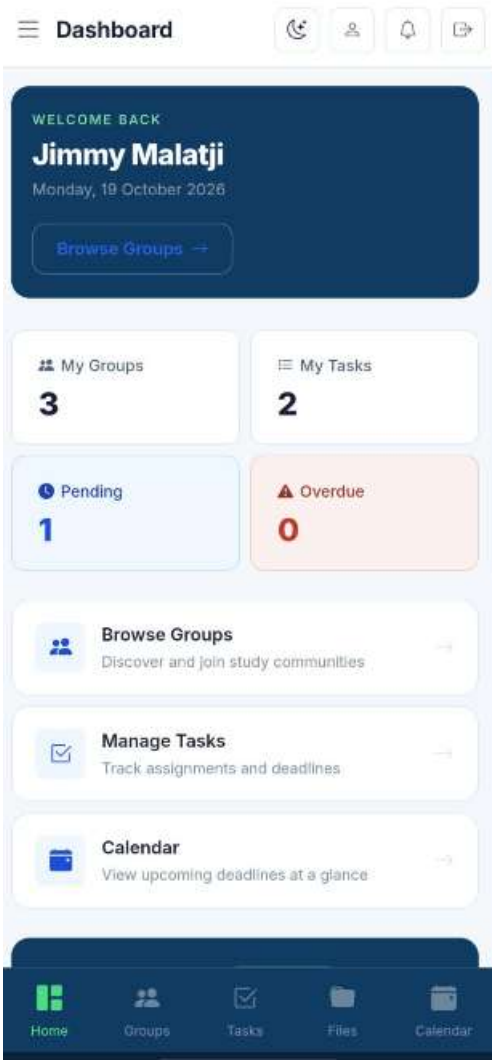
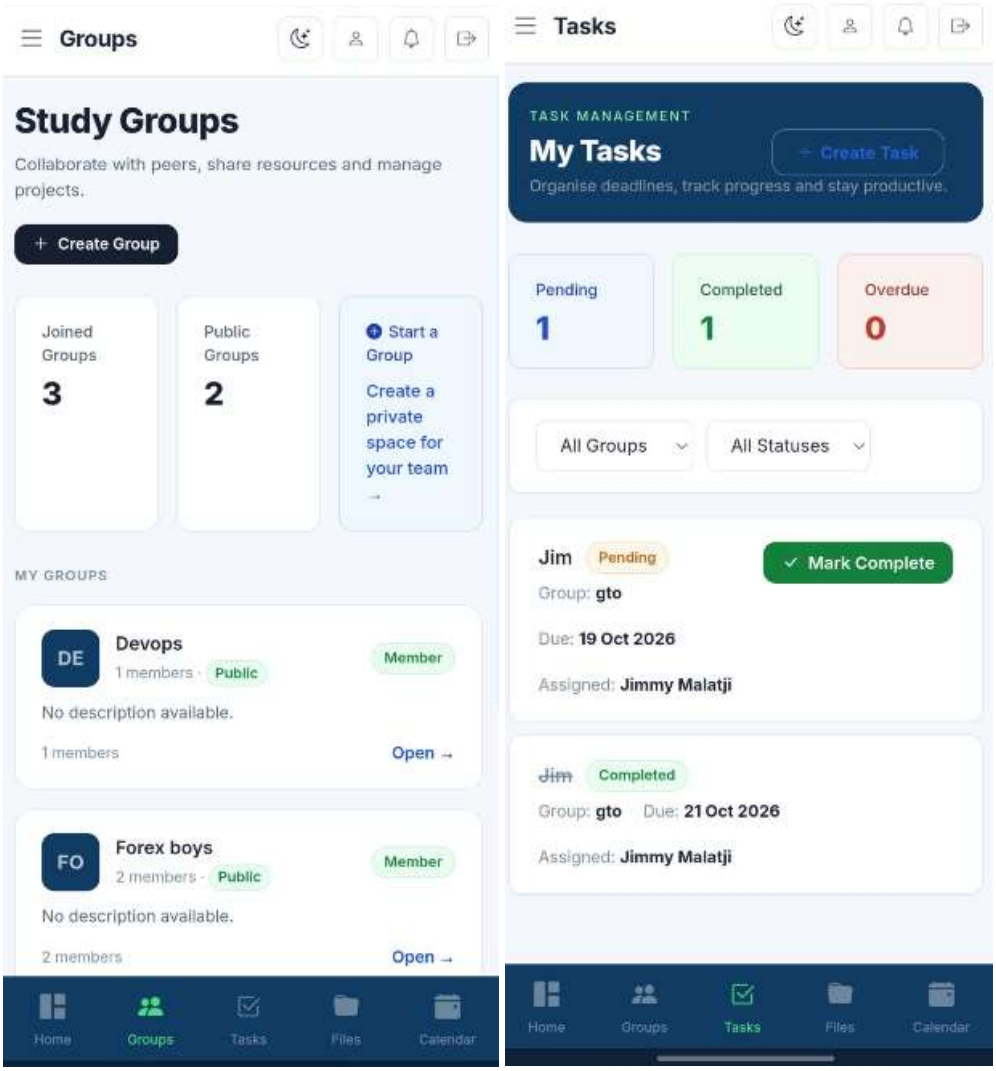


Figure 13. Primary mobile wireframes for login home and groups.

The native welcome screen precedes the illustrated web login page. Create an account and Sign in open their shared web routes; Help opens support. Google temporarily opens the system browser before returning to the app. After authentication, home shortcuts or responsive bottom navigation lead to Groups and the other tools. Android Back follows the previous page.

Group Tasks Files and Calendar



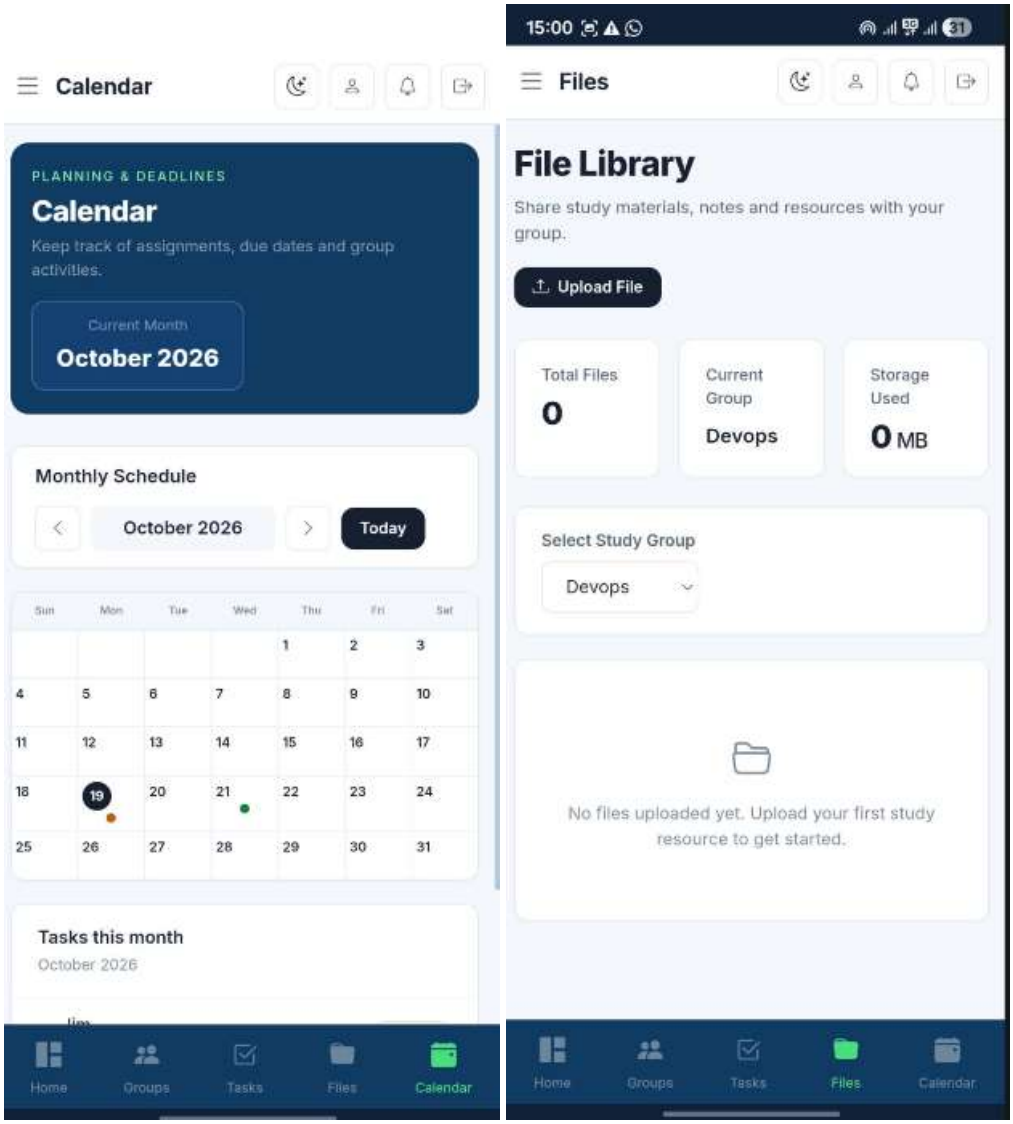


Figure 14. Primary mobile wireframes for group tasks files and calendar.

Selecting a group retains context across Overview, Members, Files and Chat. Global Tasks/Files can filter by joined group, and Calendar displays monthly task deadlines. Back follows history. The new system file picker returns a selection to the current form; the save picker returns download feedback. The original four-screen image is retained as a design reference, not a screenshot proving v1.3 device tests.

Screen Requirements and Readiness

3.5 Screen Level Requirements and Annotations

Screen	Required Inputs / Actions	Expected Feedback
Login / Register	Email, password, account fields, sign-in/register action.	Validation messages, loading state, success redirect, failed-login message.
Dashboard	Open group/task/calendar shortcuts; view summaries.	Current counts, upcoming deadlines, recent activity, clear empty state.
Study Groups	Search, create, join, open group.	Search results, request status, created-group confirmation.
Group Workspace	Switch Overview/Members/Files/Chat; manage members where authorised; upload files and chat. Use global Tasks for task actions.	Updated lists, permission messages, success/error alerts.
Tasks	Filter by group/status, create a self-assigned task and mark own task done.	Due-date validation, updated progress and completion state.
Files	Choose group, select file, upload, download.	File-size/type validation, upload progress and download result.
Calendar	Navigate months and view task due dates from joined groups.	Monthly task markers and displayed due dates.
Administration	View system totals and activate/deactivate other accounts.	Protected access, permission checks and account-status feedback.

3.6 Development Readiness Checklist

- Every high-priority requirement maps to a screen, navigation path and acceptance criterion.
- Forms identify required fields, validation and feedback.
- Website and Android labels and role permissions agree with the separate maps.
- Database entities support accounts, groups, memberships, requests, invitations, tasks, files, chat, notifications and provider sign-in records.
- Test evidence covers each labelled navigation path, including denied access, failed sign-in, expired invitations and Android retry.
- Diagram and requirement changes are reviewed before implementation.

This Task 1 submission retains the project plan and feasibility material, provides separate website and Android navigation diagrams, and connects screen layouts to the current collaboration workflows.

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Appendix A — GitHub Actions Workflow Evidence

These supplied screenshots record Android CI build/test execution and artifact publication, not a physical-phone test or website-backend test. The workflow list includes later runs at commit 9c03104; the detailed run #1 and artifact screenshots show the earlier 08d3bcc build. Do not label all four as the v1.3 test run.

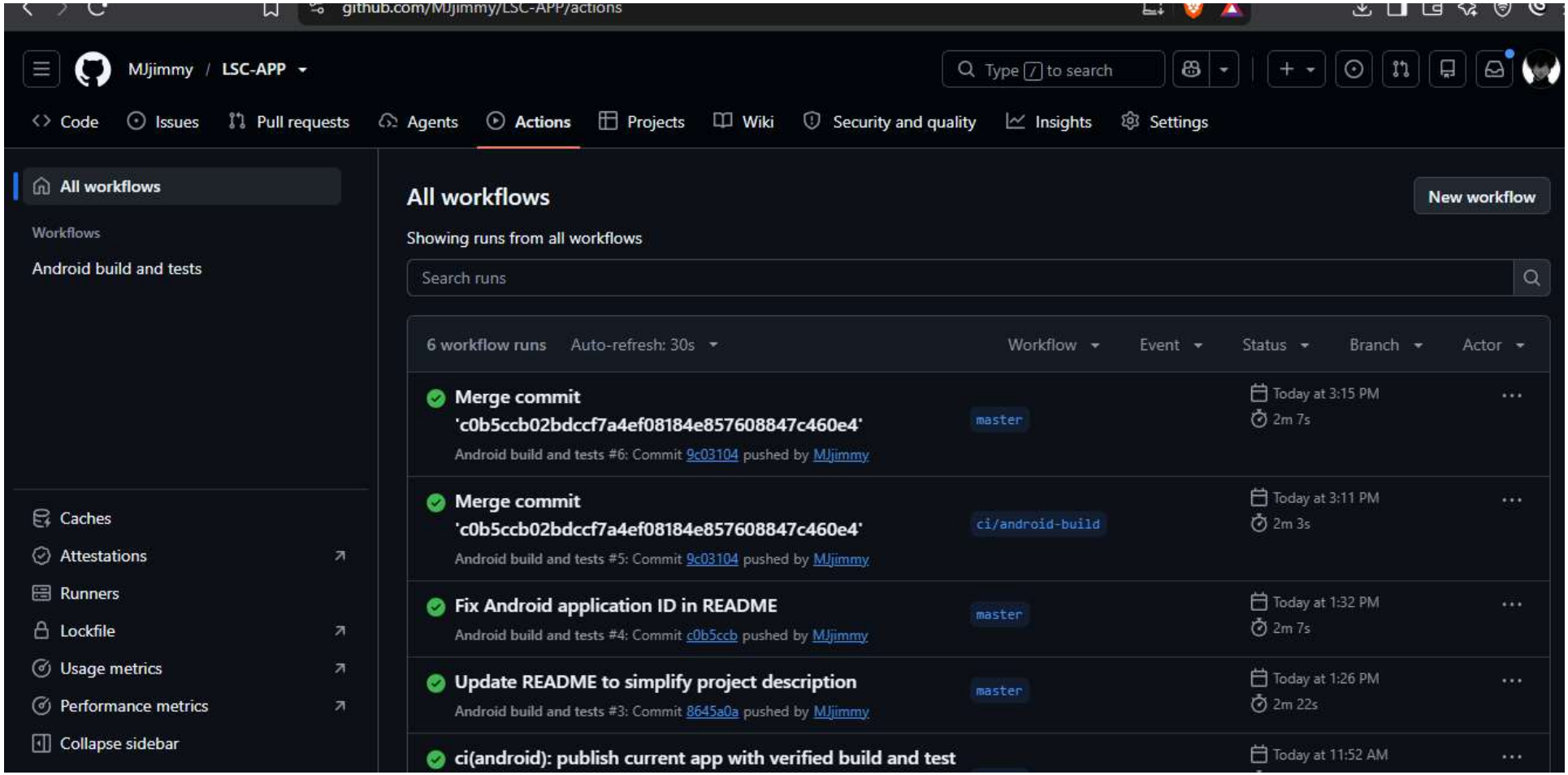


Figure 15. Android Actions workflow history: successful runs on master and ci/android-build, including later commit 9c03104.

Android workflow job and completed steps

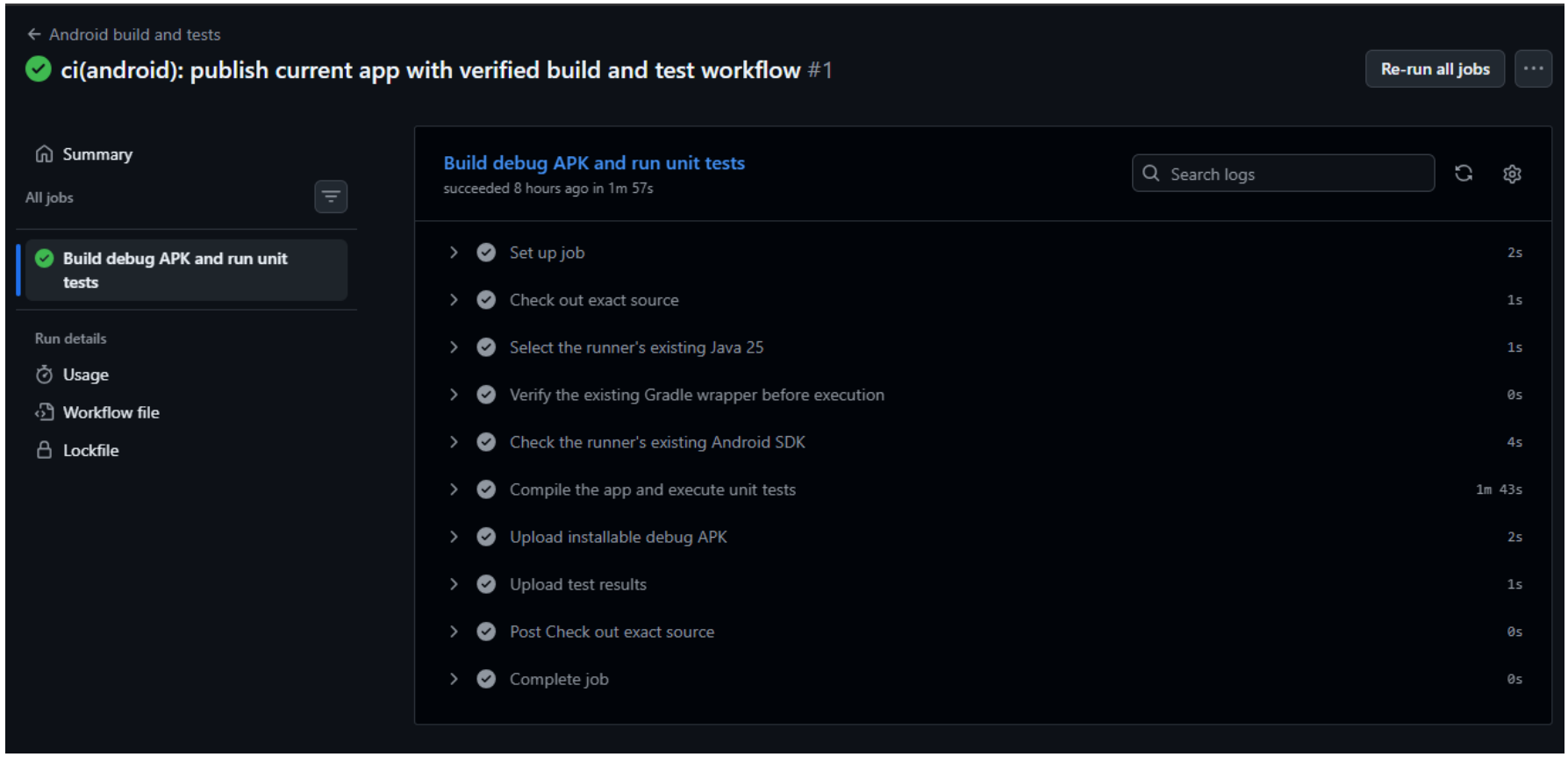


Figure 16. Earlier workflow run #1: source checkout, Java/SDK and Gradle-wrapper checks, compilation/unit tests, APK upload and test-result upload completed successfully.

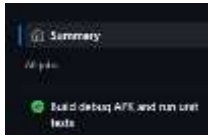


Figure 17. Supplied close-up of the successful Build debug APK and run unit tests job; this is a status detail, not an additional independent test run.

Workflow summary and published artifacts

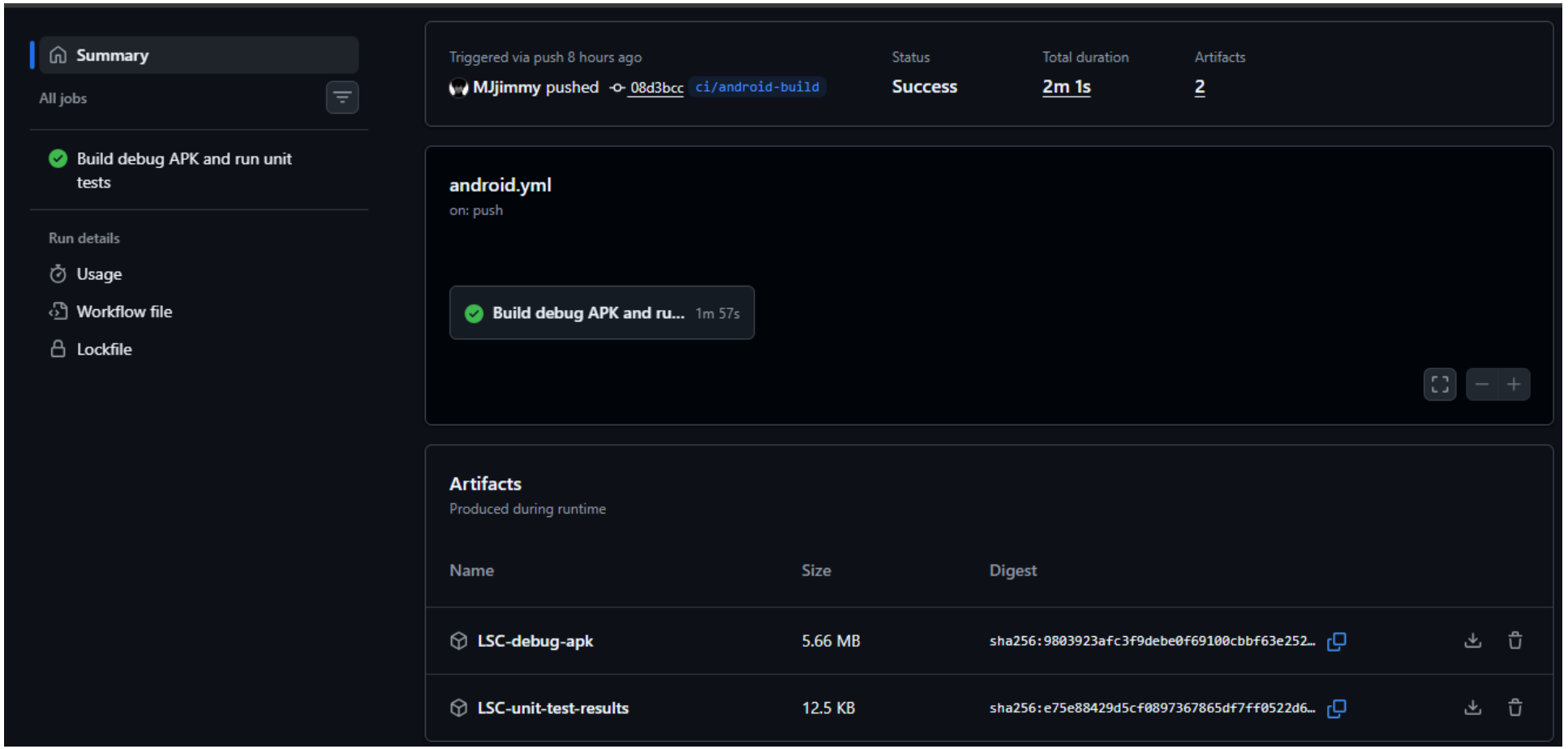


Figure 18. Earlier 08d3bcc / ci/android-build run: Success, total duration 2m 1s, and published LSC-debug-apk and LSC-unit-test-results artifacts. Artifact publication does not itself prove installation or device feature tests.